

Pair-State Consistency in Paired Combinatorial Logit Assortment Planning

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Abstract. Assortment planning under the paired combinatorial logit (PCL) model couples a display decision with demand substitution across product pairs. When one member of a pair is offered and the other is not, the pair remains active and contributes the attraction weight of the offered product. We examine two incomplete formulations that replace the full pair-state system with either one global singleton term per offered product or terms for jointly offered pairs only. The PCL choice system of [Koppelman and Wen \(2000\)](#) and the assortment formulation of [Zhang et al. \(2020\)](#) provide the reference model. On a six-product instance with a display limit of four, the unique optimal assortment is $\{3, 5, 6\}$ with expected margin 0.323647238294198. Both incomplete formulations select $\{2, 3, 5, 6\}$. Under the reference model, that assortment has expected margin 0.285975423739680, which is 11.639776% below optimum. Omitting pair states changes both the choice probabilities and the resulting assortment decision.

Keywords. assortment optimization; paired combinatorial logit; discrete choice; revenue management; mixed-integer linear programming

1. Problem Motivation

Business-level description. A retailer must choose up to four products for a limited display. The demand science team has estimated an attraction weight and a normalized contribution margin for each product. It has also calibrated a paired combinatorial logit choice model with a dissimilarity parameter for every unordered product pair and an outside-option weight for customers who leave without purchasing. The retailer selects the assortment that maximizes expected contribution margin while accounting for substitution among the displayed products.

A display limit requires the retailer to compare discrete assortments whose products compete for demand. The choice model must therefore map every offered set to the corresponding purchase probabilities before the assortments can be ranked.

The paired combinatorial logit (PCL) model of [Koppelman and Wen \(2000\)](#) assigns a separate choice nest to each unordered pair of products. [Zhang et al. \(2020\)](#) study assortment optimization under this demand model. We adopt the standard PCL choice system and use an exact linear-fractional reformulation as the reference model.

Pairwise demand creates a state-consistency requirement. Each pair contributes differently when only its first product, only its second product, or both products are offered. A formulation that omits or merges these states evaluates an altered demand model and may select a different assortment.

2. Problem Definition

Let $n \geq 2$ be the number of products, let $N = \{1, \dots, n\}$ be the product set, and let $\mathcal{P} = \{(i, j) \in N \times N \mid i < j\}$ be the set of unordered product pairs. The retailer may offer at most K products, where $1 \leq K \leq n$. Let $x = (x_i)_{i \in N}$ be the offer vector. Binary variable x_i equals one when product $i \in N$ is offered and zero otherwise, so a feasible assortment satisfies $\sum_{i \in N} x_i \leq K$.

Product $i \in N$ has attraction weight $v_i > 0$ and normalized contribution margin $r_i \geq 0$. The outside option has attraction weight $v_0 > 0$. Pair $(i, j) \in \mathcal{P}$ has dissimilarity parameter $\gamma_{ij} \in (0, 1]$. We use one choice nest for each pair in $\mathcal{P}$, which fixes the scale of v_0 .

For any feasible assortment x , the PCL model determines the purchase probabilities of the offered products and the outside option. The retailer chooses x to maximize the resulting expected contribution margin.

3. Reference Formulation

Following [Koppelman and Wen \(2000\)](#) and [Zhang et al. \(2020\)](#), define the inclusive value of pair $(i, j) \in \mathcal{P}$ as

$$I_{ij} = \left(v_i^{1/\gamma_{ij}} + v_j^{1/\gamma_{ij}} \right)^{\gamma_{ij}}. \quad (1)$$

When both products are offered, their contributions to the pair nest are

$$\mu_{ij}^i = I_{ij} \frac{v_i^{1/\gamma_{ij}}}{v_i^{1/\gamma_{ij}} + v_j^{1/\gamma_{ij}}}, \quad (2)$$

$$\mu_{ij}^j = I_{ij} \frac{v_j^{1/\gamma_{ij}}}{v_i^{1/\gamma_{ij}} + v_j^{1/\gamma_{ij}}}. \quad (3)$$

Let $S = \{10, 01, 11\}$ be the set of active pair states. For each $(i, j) \in \mathcal{P}$, binary variables y_{ij}^{10} , y_{ij}^{01} , and y_{ij}^{11} indicate that only product i , only product j , or both products are offered. When neither product is offered, all three state variables equal zero.

For state $s \in S$, let b_{ij}^s and a_{ij}^s denote the denominator and expected-margin numerator coefficients. The denominator coefficients are $b_{ij}^{10} = v_i$, $b_{ij}^{01} = v_j$, and $b_{ij}^{11} = I_{ij}$. The numerator coefficients are $a_{ij}^{10} = r_i v_i$, $a_{ij}^{01} = r_j v_j$, and $a_{ij}^{11} = r_i \mu_{ij}^i + r_j \mu_{ij}^j$.

The PCL denominator and expected-margin numerator are

$$D(x) = v_0 + \sum_{(i,j) \in \mathcal{P}} \sum_{s \in S} b_{ij}^s y_{ij}^s, \quad (4)$$

$$R(x) = \sum_{(i,j) \in \mathcal{P}} \sum_{s \in S} a_{ij}^s y_{ij}^s. \quad (5)$$

The expected contribution margin is $R(x)/D(x)$. An exactly-one-offered pair contributes v_i or v_j to Eq. (4) and the corresponding margin-weighted term to Eq. (5). A pair vanishes only when neither product is offered.

We apply the Charnes–Cooper transformation of [Charnes and Cooper \(1962\)](#). Let $t = 1/D(x)$ and $w_{ij}^s = ty_{ij}^s$ for every $(i, j) \in \mathcal{P}$ and $s \in S$. Define

$$\bar{D} = v_0 + \sum_{(i,j) \in \mathcal{P}} \max_{s \in S} b_{ij}^s, \quad \underline{t} = \frac{1}{\bar{D}}, \quad \bar{t} = \frac{1}{v_0}. \quad (6)$$

The inequalities $v_0 \leq D(x) \leq \bar{D}$ imply $0 < \underline{t} \leq t \leq \bar{t}$. The resulting mixed-integer linear program is

$$\max \sum_{(i,j) \in \mathcal{P}} \sum_{s \in S} a_{ij}^s w_{ij}^s, \quad (7)$$

$$\text{subject to } v_0 t + \sum_{(i,j) \in \mathcal{P}} \sum_{s \in S} b_{ij}^s w_{ij}^s = 1, \quad (8)$$

$$\sum_{i \in N} x_i \leq K, \quad (9)$$

$$y_{ij}^{11} \leq x_i, \quad y_{ij}^{11} \leq x_j, \quad y_{ij}^{11} \geq x_i + x_j - 1 \quad \forall (i, j) \in \mathcal{P}, \quad (10)$$

$$y_{ij}^{10} = x_i - y_{ij}^{11}, \quad y_{ij}^{01} = x_j - y_{ij}^{11} \quad \forall (i, j) \in \mathcal{P}, \quad (11)$$

$$\underline{t} y_{ij}^s \leq w_{ij}^s \leq \bar{t} y_{ij}^s \quad \forall (i, j) \in \mathcal{P}, s \in S, \quad (12)$$

$$t - \bar{t}(1 - y_{ij}^s) \leq w_{ij}^s \leq t - \underline{t}(1 - y_{ij}^s) \quad \forall (i, j) \in \mathcal{P}, s \in S, \quad (13)$$

$$x_i \in \{0, 1\} \quad \forall i \in N, \quad (14)$$

$$y_{ij}^s \in \{0, 1\}, \quad w_{ij}^s \geq 0 \quad \forall (i, j) \in \mathcal{P}, s \in S, \quad (15)$$

$$\underline{t} \leq t \leq \bar{t}. \quad (16)$$

Constraints (10) and (11) select the unique state of every active pair. Constraints (12) and (13) impose $w_{ij}^s = ty_{ij}^s$. The normalization in Eq. (8) converts the fractional objective into the linear objective in Eq. (7). The resulting variables describe one assortment and its associated PCL choice probabilities.

4. Failure of Incomplete Pair States

Two incomplete formulations replace the three-state system with restricted pair-state representations. Using superscript SP for singleton plus pair, the first formulation uses

$$D^{\text{SP}}(x) = v_0 + \sum_{i \in N} v_i x_i + \sum_{(i,j) \in \mathcal{P}} I_{ij} x_i x_j, \quad (17)$$

$$R^{\text{SP}}(x) = \sum_{i \in N} r_i v_i x_i + \sum_{(i,j) \in \mathcal{P}} a_{ij}^{11} x_i x_j. \quad (18)$$

The global singleton term counts each offered product once, while the reference model contributes one exactly-one-offered term for every absent counterpart.

Using superscript JP for jointly offered pairs only, the second formulation uses

$$D^{\text{JP}}(x) = v_0 + \sum_{(i,j) \in \mathcal{P}} I_{ij} x_i x_j, \quad (19)$$

$$R^{\text{JP}}(x) = \sum_{(i,j) \in \mathcal{P}} a_{ij}^{11} x_i x_j. \quad (20)$$

Every pair with exactly one offered member disappears from this choice system. Both formulations therefore change the numerator and denominator used to rank assortments.

The six-product instance is derived from the computational artifact associated with exact PCL assortment optimization under general linear constraints ([Feng et al., 2026a,b](#)). Product IDs are remapped to $N = \{1, \dots, 6\}$, the display limit is $K = 4$, and $v_0 = 3.84538427435401$. The outside-option probability equals 0.25 when all six products are offered. Table 1 compares the exact reference solution with the decisions of the two incomplete formulations.

Table 1. Assortments selected under the reference and incomplete formulations.

Formulation	Selected products	Objective under the formulation
Reference PCL formulation	{3, 5, 6}	0.323647238294198
Singleton-plus-pair formulation	{2, 3, 5, 6}	0.245658160940
Jointly-offered-pairs-only formulation	{2, 3, 5, 6}	0.188961826760

Each incomplete formulation is solved to optimality under its own objective. The two altered choice systems select the same four-product assortment.

Illustrative instance. The exact mixed-integer linear program and complete enumeration of all 57 feasible assortments identify the unique optimum {3, 5, 6} with expected margin 0.323647238294198. Both incomplete formulations select {2, 3, 5, 6}. Under the reference PCL model, that assortment has expected margin 0.285975423739680, giving absolute regret 0.037671814554518 and relative regret 11.639776%.

The display-capacity constraint holds for all three decisions. The discrepancy follows from the choice probabilities assigned to exactly-one-offered pairs. Preserving the complete pair-state system is therefore necessary for the optimization model to evaluate the calibrated PCL demand model.

Supporting materials for the case study, including the derivative data, source document, reference solver, and verification instructions, are available in the [corresponding folder of the OR-Bench GitHub repository](#).

References

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